

COS 135 HW: 1

C the sights (10pts)

Homework Tasks:

- Create a C program named “shapes”. [Don’t forget to comment your code as you go]
- In shapes.c have the following:

1. A function called “**createPyramid**” that takes in a single integer. It will print off to the screen a pyramid of that height when called. As an example:

```
createPyramid(5);
```

Should create...

```
  x
 xxx
xxxxx
xxxxxxx
xxxxxxxxx
```

```
createPyramid(3);
```

Should create...

```
  x
 xxx
xxxxx
```

2. a function called “**createPyramidReverse**” that takes a single integer as an argument. It will print off to the screen an upside down version of the previous function.

3. “**gradeHistogram**” that takes 2 parameters. An array of integers and a count of how many things are in the array.

This function should show a rudimentary histogram of the numbers on a 0-100 scale based on the grade.

A (90+):	xxxxxxxxxx	(10 x's)
B (80-89):	xxxxxxx	(8 x's)
C (70-79):	xxxxxx	(6 x's)
D (60-69):	xxxx	(4 x's)
F (<60):	xx	(2 x's)

As an example:

```
gradeHistogram([90,95,50,72,85],5);
```

should print off...

```
xxxxxxxxxx
xxxxxxxxxx
xx
xxxxxx
xxxxxx
```

- Have the program execute the following three lines of code in your main function:
 createPyramidReverse(7);
 createPyramid(7);
 gradeHistogram([90,64,50,72,85,95],6);

Attach this c file to the brightspace submission

Rubric

The function “createPyramid” works	2pts
The function “createPyramidReverse” works	2pts
The function “gradeHistogram” works	4pts
Everything named how it should be	1pt
Code was commented	1pt